Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

1. Suppose that we choose to track targets using NCV models. A candidate track has passed the persistence test and we desire to initiate the execution of a Kalman filter to track the target.

1 / 1 point

The most recent position measurements have been: $z_0 = 3$, $z_{-1} = 2$, $z_{-2} = -1$, and $z_{-3} = -6$. If $\Delta t = 0.5$ seconds, what is the velocity estimate to use for $\hat{x}_0$ when initializing the filter?

2

Correct
Yes. This is correct.

2. Suppose that we have detected exceedances in a 3×3 frame of data (the "track gate" comprises this entire frame):

1 / 1 point

$E_{m,n}$	$m = 1$	$m = 2$	$m = 3$
$n = 1$	0	10	2
$n = 2$	0	5	1
$n = 3$	0	1	1

What is the centroid of these exceedances in the horizontal direction?

2.2

Correct
Yes. That is correct.

3. For this question, you will use results from the Coursera Labs Jupyter notebook for Lesson 2.4.2.

1 / 1 point

First, change the true radius to the target to `r = 10000`. Then, execute the lab. What is the true y-coordinate of the target location? Enter your answer with one digit to the right of the decimal point.

7071.1

Correct
Yes. That is correct.

4. For this question, you will once again use the Coursera Labs Jupyter notebook from Lesson 2.4.2. As with the prior question, change the true radius to the target to `r = 10000`. Then, what is the corrected estimate of the y-coordinate of the target location? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

7071.8

Correct
Yes. That is correct.

5. In the IMM Kalman filter, what is the interpretation of the p_{ij} matrix?

1 / 1 point

- ○ It is the conditional probability of arriving to mode j at time k from mode i.
- ○ It is the pmf describing the probability that the target is operating in each mode at time k.
- ◉ It is the probability of transitioning from mode i to mode j between timesteps.
- ○ It is the overall state estimate at time k.

Correct
Yes. This is correct.

6. In the IMM Kalman filter, what is the interpretation of the $\mu_{i|j,k-1}$ matrix?

1 / 1 point

- ◉ It is the conditional probability of arriving to mode j at time k from mode i.
- ○ It is the overall state estimate at time k.
- ○ It is the pmf describing the probability that the target is operating in each mode at time k.
- ○ It is the probability of transitioning from mode i to mode j between timesteps.

Correct
Yes. This is correct.

7. Run the IMM code in the Coursera Labs Jupyter notebook for Lesson 2.4.4. What does the IMM identify as the most likely target mode at time $t = 75$ seconds?

1 / 1 point

- ○ The NCP mode.
- ◉ The NCV mode.

Correct
No. Refer to Lesson 2.4.4.

8. Change the initial random number "seed" in the Coursera Labs Jupyter notebook for Lesson 2.4.4 by replacing `randn("seed",5)` with `randn("seed",1)`. Then, re-run the code. After making this change, what is the percentage of time that the error is outside of the confidence bounds? Enter your answer with one digit to the right of the decimal point.

1 / 1 point

0.5

Correct
Yes. That is correct.

9. For the final two questions in this quiz, let $\sigma_{\tilde{w}} = 0.1$, $\Sigma_{\tilde{v}} = 0.2$, and $\Delta t = 0.5$.

1 / 1 point

What is the value of λ for an α-β filter? Enter your answer with four digits to the right of the decimal point.

0.0559

Correct
Yes. That is correct.

10. What is the value of β for an α-β filter? Enter your answer with four digits to the right of the decimal point.

1 / 1 point

0.0473

Correct
Yes. That is correct.